Type MS · Writing the equation of a line through two points

Answer 1 of 12 · Question 8 part a · 3 marks

2023 JUN · Paper 02 · extract 1 of 5 · find part a in this mark scheme extract

Question	Scheme	Marks
8(a)	$(\text{Grad}_{AB}) = \frac{2-8}{12-(-6)}$ oe $y-2 = "\left(-\frac{1}{3}\right)"(x-12)$ oe $\Rightarrow x+3y-18=0$ or $-x-3y+18=0$ oe	M1 dM1 (A1 on ePen) A1 [3]
(b)	Length $= \sqrt{(12--6)^2+(2-8)^2} = 6\sqrt{10}$ eg $\sqrt{360}$ oe	M1A1 [2]
(c)	$\left(\left(\frac{2\times-6+1\times12}{1+2}, \frac{2\times8+1\times2}{1+2}\right)\right)$ $(0,6)$ or $m=0, n=6$	B1 B1 (M1A1 on ePen) [2]
(d) Mark parts (i) and (ii) together	Gradient of $CA = \frac{q-8}{p+6}$ **or** gradient of $CB = \frac{q-2}{p-12}$ oe $\frac{q-8}{p+6} = -\frac{1}{\frac{q-2}{p-12}} = -\left(\frac{p-12}{q-2}\right) \Rightarrow q^2-10q+16 = -p^2+6p+72$ Gradient of $XC = -\frac{1}{"-\frac{1}{3}"} \Rightarrow -\frac{1}{"-\frac{1}{3}"} = \frac{q-6}{p-0} (\Rightarrow q = 3p+6)$ $('3p+6')^2-10('3p+6')+16 = -p^2+6p+72 \Rightarrow 10p^2-80=0$ $10p^2-80=0 \Rightarrow p=\sqrt{8}$ oe $q = 3\times\sqrt{8}+6 = 6+6\sqrt{2}$ oe	M1 M1 B1ft ddM1A1 M1A1 [7]
ALT1 Mark parts (i) and (ii) together	$(\text{midpoint of } AB =)\left(\frac{-6+12}{2}, \frac{8+2}{2}\right)$ $(=(3,5))$ $(\text{radius of } C) = \frac{"6\sqrt{10}"}{2}$ oe or $\sqrt{(-6-3)^2+(8-5)^2}$ oe $y = \frac{-1}{"-\frac{1}{3}"}x+6$ $(y=3x+6)$ oe $(x-"3")^2+(y-"5")^2 = \left(\frac{"6\sqrt{10}"}{2}\right)^2 \Rightarrow (x-"3")^2+("3x+6"-"5")^2 = \left(\frac{"6\sqrt{10}"}{2}\right)^2$ $10x^2=80$ oe $x=\sqrt{8}$ oe $p=\sqrt{8}, q=6+3\sqrt{8}$ oe	M1 M1 B1ft ddM1 A1 M1 A1 [7]

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ALT2 Mark parts (i) and (ii) together	$\left((AC)^2=\right)(p--6)^2+(q-8)^2$ or $(AC=)\sqrt{(p--6)^2+(q-8)^2}$ oe	M1
	$\left((BC)^2=\right)(p-12)^2+(q-2)^2$ or $(AC=)\sqrt{(p-12)^2+(q-2)^2}$ oe	M1
	$q=3p+6$	B1ft
	$\left((AB)^2=(AC)^2+(BC)^2\right)$	
	$\left("6\sqrt{10}"\right)^2=(p--6)^2+("3p+6"-8)^2+(p-12)^2+("3p+6"-2)^2$	ddM1A1
	$10p^2=80$ oe	
	$p=\sqrt{8}$ oe	M1 A1
	$p=\sqrt{8}, q=6+3\sqrt{8}$ oe	[7]
(e)	Length $CX=\sqrt{\left("\sqrt{8}"-"0"\right)^2+\left("6+6\sqrt{2}"-"6"\right)^2}\left(=4\sqrt{5}\right)$	M1
	Area of triangle $ABC=\frac{1}{2}\times"4\sqrt{5}"\times"6\sqrt{10}"=60\sqrt{2}$ oe	dM1 A1 [3]
ALT	$\frac{1}{2}\begin{vmatrix}-6 & 12 & "\sqrt{8}" & -6\\ 8 & 2 & "6+3\sqrt{8}" & 8\end{vmatrix}$ oe	M1
	$\frac{1}{2}\left[\left(-6\times2+12\times"(6+3\sqrt{8})"+"\sqrt{8}"\times8\right)-\left(-6\times"(6+3\sqrt{8})"+"\sqrt{8}"\times2+12\times8\right)\right]$ oe	dM1
	$60\sqrt{2}$ oe	A1 [3]
		Total 17 marks

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Part	Mark	Notes
(a)	M1	For correctly finding the gradient of AB in unsimplified form.
	dM1 (A1 on ePen)	For a full and correct attempt to find the equation of the line using their gradient. No simplification is required. If using $y = mx + c$, a value for c must be found.
	A1	For a correct equation in the required form.
(b)	M1	For using a correct method to find the length of line segment AB, in unsimplified form.
	A1	For the correct exact length.
(c)	B1 (M1 on ePen)	For either coordinate correct.
	B1 (A1 on ePen)	For both correct coordinates (0, 6) For part c the values of m and n can be explicitly identified or written in a coordinate.
In part (d) allow p to be interchangeable with x, q to be interchangeable with y throughout		
(d)	M1	For a correct statement of the gradient for either CA or CB
	M1	For using the negative perpendicular of either gradient and equating the gradients to form an equation in terms of p and q only.
	B1ft	For finding the negative reciprocal of their gradient of XC and placing this equal to a correct expression in p and q as shown.
	ddM1	For correctly substituting their **linear** expression for p or q into a **quadratic** equation in q or p to obtain an equation in one variable. Must use their $q = 3p + 6$ and is dependent on both previous method marks.
	A1	For the correct two term quadratic..
	M1	For **correctly solving their** quadratic to find a value for either p or q
	A1	For **both** p and q correct.
ALT1 (d)	M1	For the correct method to find the midpoint of AB
	M1	For the correct method to find the radius of C, ft their answer from part b if used.
	B1ft	For the equation of the line, unsimplified, ft their gradient of AB
	ddM1	For correctly substituting their $y = 3x + 6$ into the equation of a circle, using their midpoint of AB and their radius of C Must use their $y = 3x + 6$ and is dependent on both previous method marks.
	A1	Correct equation
	M1	For **correctly solving their** quadratic to find a value for either x or y
	A1	For **both** p and q correct.
ALT2 (d)	M1	For the correct method to find the length of AC or $(AC)^2$
	M1	For the correct method to find the length of BC or $(BC)^2$
	B1ft	For the equation of the line, unsimplified, ft their gradient of AB
	ddM1	For correctly substituting their lengths and their $q = 3p + 6$ into a correct Pythagorean equation, dependent on both previous method marks.
	A1	Correct equation
	M1	For **correctly solving their** quadratic to find a value for either p or q
	A1	For **both** p and q correct.

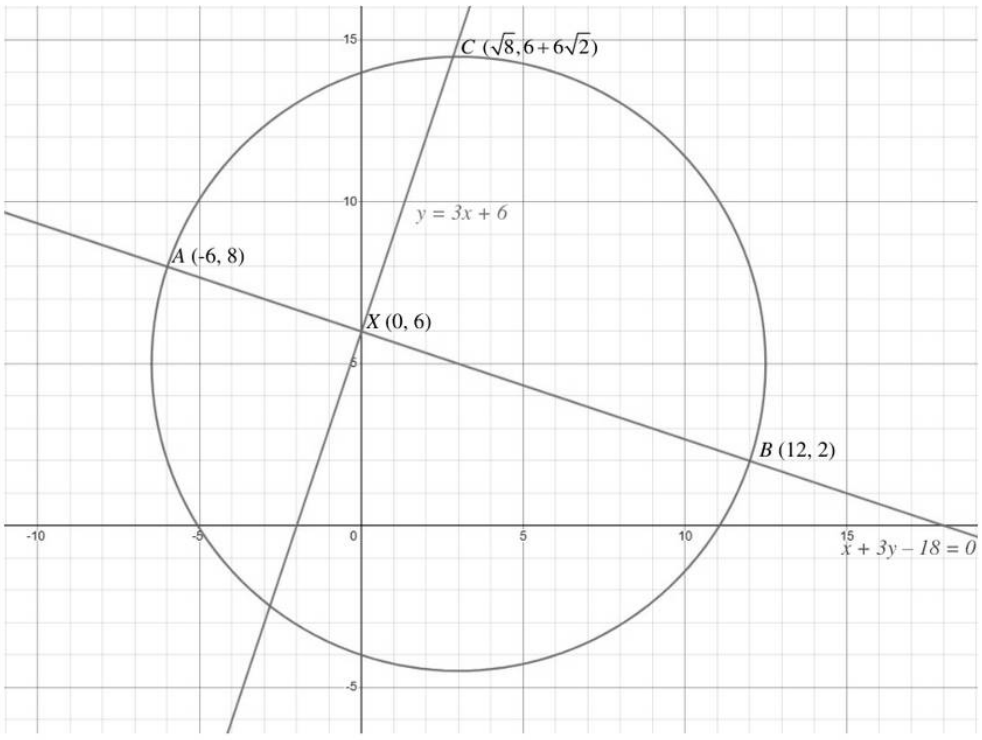
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(e)	**M1**	For using a correct method to find the length of the perpendicular from AB to C
	dM1	For using their results from part (b) and their length of perpendicular from AB to C with the correct formula for the area of a triangle. Dependent on previous method mark.
	A1	For the correct area of $60\sqrt{2}$ or $30\sqrt{8}$ (units^2)
ALT	**M1**	For a correct statement for the area such as the one shown, using their values of p and q
	dM1	For the correct evaluation of their determinant Dependent on previous method mark.
	A1	For the correct area of $60\sqrt{2}$ or $30\sqrt{8}$ (units^2)

Useful Sketch for Parts c/d – look for any working on or near a sketch.



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Question	Scheme	Marks

Type MS · Writing the equation of a line through two points

Answer 2 of 12 · Question 11 part a · 2 marks

2019 JUN · Paper 01 (regional R) · extract 1 of 4 · find part a in this mark scheme extract

Number		
11 (a)	$\frac{y-6}{x-5}=\frac{6-3}{5--1}\Rightarrow 2y-12=x-5$	M1A1 [2]
(b)	$\left(\frac{2\times5+1\times-1}{2+1},\frac{2\times6+1\times3}{3}\right)\Rightarrow(3,\ 5)$*	M1A1 [2]
(a)		
M1	A fully correct method for finding the equation of a straight line e.g. $\frac{y-y_1}{x-x_1}=\frac{y_2-y_1}{x_2-x_1}$	
A1	$2y-12=x-5$ oe	
(b)		
M1	Use of $\left(\frac{qx_1+px_2}{p+q},\frac{qy_1+py_2}{p+q}\right)$ or $\begin{pmatrix}-1\\3\end{pmatrix}+\frac{2}{3}\times\begin{pmatrix}6\\3\end{pmatrix}$	
A1	Obtains the **given** coordinates	

Type MS · Writing the equation of a line through two points

Answer 2 of 12 · Question 11 part a · 2 marks

2019 JUN · Paper 01 (regional R) · extract 2 of 4 · find part a in this mark scheme extract

(c)	Gradient of perpendicular to AB = -2	B1
	$\frac{5-n}{3-m}=-2\Rightarrow(11=2m+n)$	M1
	Radius = 5, so the length of AC = 10	
	$100=(3-n)^2+(-1-m)^2\Rightarrow(90=n^2-6n+m^2+2m)$	M1
	Solves simultaneous equations $90=121-44m+4m^2-6(11-2m)+m^2+2m$	depM1
	$35=5m^2-30m\Rightarrow m^2-6m-7=0$	depM1
	$(m-7)(m+1)=0\Rightarrow m=7,\ n=-3$	A1 [6]
ALT (c) using vectors		
(c)	$\overrightarrow{AB}=\begin{pmatrix}5\\6\end{pmatrix}-\begin{pmatrix}-1\\3\end{pmatrix}=\begin{pmatrix}6\\3\end{pmatrix}\Rightarrow$ Perpendicular vector to $\overrightarrow{AB}=\begin{pmatrix}3\\-6\end{pmatrix}$	B1
	$\lvert\overrightarrow{AB}\rvert=\sqrt{45}=3\sqrt{5}\Rightarrow\lvert AP\rvert=2\sqrt{5}\qquad\lvert\overrightarrow{AC}\rvert=10$	M1M1
	Using Pythagoras $\lvert\overrightarrow{PC}\rvert=\sqrt{100-20}=\sqrt{80}=4\sqrt{5}$	depM1
	$\overrightarrow{PC}=\frac{4\sqrt{5}}{3\sqrt{5}}\times\begin{pmatrix}3\\-6\end{pmatrix}=\begin{pmatrix}4\\-8\end{pmatrix}$	depM1
	$\Rightarrow$ Coordinates of C are $(3+(4),\ 5+(-8))\Rightarrow(7,-3)$	A1 [6]

(c)

B1 Gradient of perpendicular to AB = -2 (Can be implied by M1)

M1 Obtains an equation using the gradient of the perpendicular and the points (m, n) and P (Condone if given in terms of x and y)

M1 Obtains a second equation using AC = 10 and the points A and C

depM1 Solve simultaneously to obtain a 3 term quadratic

depM1 Solve their 3 term quadratic

A1 $m = 7,\ n = -3$

ALT:

B1 Perpendicular vector to $\overrightarrow{AB}=\begin{pmatrix}3\\-6\end{pmatrix}$

M1 Finds $\lvert\overrightarrow{AB}\rvert$ or $\lvert\overrightarrow{AP}\rvert$ or $\overrightarrow{PC}=\begin{pmatrix}3\\5\end{pmatrix}+\lambda\begin{pmatrix}3\\-6\end{pmatrix}$

M1 States $\lvert\overrightarrow{AC}\rvert=10$ or $\lambda=\frac{4}{3}$

depM1 Use of Pythagoras to find $\lvert\overrightarrow{PC}\rvert$ or $\overrightarrow{PC}=\begin{pmatrix}3\\5\end{pmatrix}+\frac{4}{3}\times\begin{pmatrix}3\\-6\end{pmatrix}$

depM1 Finds $\overrightarrow{PC}$

A1 $m = 7,\ n = -3$

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Answer 2 of 12 · Question 11 part a · 2 marks

2019 JUN · Paper 01 (regional R) · extract 3 of 4 · find part a in this mark scheme extract

(d)	$\frac{y-'-3'}{x-'7'}=\frac{1}{2}\Rightarrow\left\{y=\frac{x-13}{2}\right\},\ \frac{y-3}{x--1}=-2\Rightarrow\{y=-2x+1\}$ Solving simultaneous equations by any method $\frac{x-13}{2}=-2x+1\Rightarrow p=3$ and $q=-5$	M1 M1A1 [3]
ALT (d) using vectors		
(d)	$\overrightarrow{AD}=\overrightarrow{PC}=\begin{pmatrix}4\\-8\end{pmatrix}$ Coordinates of point D $\Rightarrow(-1+(4),\ 3+(-8))=(3,-5)$	M1 M1A1 [3]
(e)	Length of AB $\sqrt{(6-3)^2+(5--1)^2}=3\sqrt{5}$ Length of CD $\sqrt{(-3--5)^2+(7-3)^2}=\sqrt{20}=2\sqrt{5}$ Area of trapezium $=\frac{1}{2}(3\sqrt{5}+2\sqrt{5})\times 4\sqrt{5}=50$ (units^2)	M1 A1 M1A1 [4]
ALT (e) using vectors		
	Area $=\frac{1}{2}\begin{vmatrix}-1 & 5 & 7 & 3 & -1\\ 3 & 6 & -3 & -5 & 3\end{vmatrix}$ $=\frac{1}{2}[(-6-15-35+9)-(15+42-9+5)]=50$	 M1A1 M1A1 [4]
		Total 17 marks

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Answer 2 of 12 · Question 11 part a · 2 marks

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Type MS · Writing the equation of a line through two points

Answer 3 of 12 · Question 2 part a · 3 marks

2022 JAN · Paper 02 (regional R) · extract 1 of 2 · find part a in this mark scheme extract

Number		
2 (a)	Gradient $= \frac{4+1}{3+7} = \frac{1}{2}$	M1
	$y+1 = \frac{1}{2}(x+7)$	M1
	$x-2y+5=0$	A1
		(3)
(b)	$AB = \sqrt{10^2+5^2} = 5\sqrt{5}$	
	$AC = \sqrt{(-7--3)^2+(7--1)^2} = 4\sqrt{5}$	M1
	$k = \frac{5}{4}$	A1
		(2)
(c)	$-2 = \frac{7-p}{-3-3}$	M1
	$12 = 7-p$	dM1
	$p=-5$	A1
		(3)
		Total 8 marks

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Answer 3 of 12 · Question 2 part a · 3 marks

2022 JAN · Paper 02 (regional R) · extract 2 of 2 · find part a in this mark scheme extract

Question number	Scheme	Marks

Type MS · Writing the equation of a line through two points

Answer 4 of 12 · Question 5 part b · 2 marks

2023 JAN · Paper 02 (regional R) · extract 1 of 3 · find part b in this mark scheme extract

5(a)	$-\frac{1}{12}=\frac{10-0}{p-123}$ $\left[\text{or } -\frac{1}{12}=\frac{0-10}{123-p}\right]$ $\Rightarrow p-123=-12\times10\Rightarrow p=3^*$	M1 A1 cso [2]
(b)	$y-10=-\frac{1}{12}(x-3)\Rightarrow 12y+x-123=0$	M1A1 [2]
(c)	$m_k=12$ $y-10=12(x-3)\Rightarrow y=12x-26$	B1 M1A1 [3]
(d)	At C: $y=12x-26$ when $y=0$, $x=\frac{26}{12}$ oe $\text{Area}_{ABC}=\frac{1}{2}\times10\times\left(123-\frac{13}{6}\right)=\frac{3625}{6}$ ALT $\text{Area}=\frac{1}{2}\begin{vmatrix}3 & 123 & 13/6 & 3\\ 10 & 0 & 0 & 10\end{vmatrix}=\frac{3625}{6}$	M1A1 M1A1 [M1A1] [4]
		Total 11 marks

Type MS · Writing the equation of a line through two points

Answer 4 of 12 · Question 5 part b · 2 marks

2023 JAN · Paper 02 (regional R) · extract 2 of 3 · find part b in this mark scheme extract

Question	Notes	Marks
5(a)	States a correct expression for the gradient in terms of p either $-\frac{1}{12}=\frac{10-0}{p-123}$ or $-\frac{1}{12}=\frac{0-10}{123-p}$ and attempts to solve their equation in p $p-123=-12\times10\Rightarrow p=...$	M1
	Finds the value of $p=3$* Must show an intermediate step e.g. $k=p\pm c$ or $c=\frac{p}{k}$ or $c=kp$	A1 cso [2]
(b)	Forms an equation using either the formula or $y=mx+c$ with the given values $y-10=-\frac{1}{12}(x-3)$ OR $y-0=-\frac{1}{12}(x-123)$	M1
	For the correct equation in the required form $12y+x-123=0$ Coefficients must be integers. Allow for $r=1, s=12, t=-123$	A1 [2]
(c)	For the gradient of the normal when $x=3$ is 12	B1
	Forms an equation of the normal using either the formula or $y=mx+c$ with their values of the gradient of the normal. $y-12=12(x-3)$	M1
	For the correct equation in the required form. $y=12x-26$	A1 [3]
(d)	Attempts to find the x coordinate at point C $y=12x-26$ when $y=0$, $x=...$ Their linear equation from (c) with $y=0$ and attempt to solve.	M1
	For the correct value of x $x=\frac{26}{12}$ oe	A1
	For attempting to find the area of triangle ABC Area $=\frac{1}{2}\times10\times\left(123-\frac{13}{6}\right)=...$ or Area $=\frac{1}{2}\begin{vmatrix}3 & 123 & 13/6 & 3\\10 & 0 & 0 & 10\end{vmatrix}$	M1
	For the correct area of the triangle $\frac{3625}{6}$ oe isw rounding	A1 [4]
		Total 11 marks

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Answer 4 of 12 · Question 5 part b · 2 marks

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Question	Scheme	Marks

Type MS · Writing the equation of a line through two points

Answer 5 of 12 · Question 9 part a · 2 marks

2024 JUN · Paper 01 (regional R) · find part a in this mark scheme extract

9(a)	$\frac{y-8}{8-3}=\frac{x-6}{6--4}\Rightarrow\frac{y-8}{5}=\frac{x-6}{10}\Rightarrow\left[y-8=\frac{1}{2}(x-6)\right]$	M1A1 [2]
(b)	$C=\left(\frac{4\times6+1\times-4}{4+1},\frac{4\times8+1\times3}{4+1}\right)=(4,7)$	B1,B1 [2]
(c)	$-2=\frac{q-7}{p-4}\Rightarrow q=15-2p$ $8\sqrt{5}=\sqrt{(4-p)^2+(7-q)^2}$ $\Rightarrow 320=(4-p)^2+(2p-8)^2\Rightarrow p^2-8p-48=0$ $\Rightarrow(p+4)(p-12)=0$ $\Rightarrow p=-4\quad q=15-2\times-4=23\Rightarrow D=(-4,23)$	M1 M1 M1 M1 A1A1 [6]
(d)	Area of triangle $ACD=\frac{1}{2}\times(23-3)\times(4--4)=80$ [square units] **ALT 1** $\frac{1}{2}\begin{bmatrix}-4 & 4 & -4 & -4\\ 3 & 7 & 23 & 3\end{bmatrix}=\frac{1}{2}\left[(-28+92-12)-(12-28-92)\right]=80$ **ALT 2** $A=\frac{1}{2}\times AC\times CD=\frac{1}{2}\times4\sqrt{5}\times8\sqrt{5}=80$	M1A1 [2] [M1A1] [M1A1]
		Total 12 marks

Type MS · Writing the equation of a line through two points

Answer 6 of 12 · Question 4 part a · 2 marks

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number		
4 (a)	Gradient $= \dfrac{11+10}{3+4} = 3$	
	$y+10=3(x+4)$ or $y-11=3(x-3)$ oe	M1A1 (2)
(b)	e.g. $\left(\dfrac{4\times-4+3\times3}{3+4}, \dfrac{4\times-10+3\times11}{3+4}\right)=(-1,-1)$	M1 A1 (2)
ALT (b)	Using Vectors	
	$\begin{pmatrix}-4\\-10\end{pmatrix}+\dfrac{3}{7}\begin{pmatrix}7\\21\end{pmatrix}=\begin{pmatrix}-1\\-1\end{pmatrix}$ or $\begin{pmatrix}3\\11\end{pmatrix}-\dfrac{4}{7}\begin{pmatrix}7\\21\end{pmatrix}=\begin{pmatrix}-1\\-1\end{pmatrix}$	{M1} {A1}
(c)	$-\dfrac{1}{3}=\dfrac{n+1}{m+1} \Rightarrow -\dfrac{1}{3}(m+1)=n+1$	M1
	$\left(\sqrt{10}\right)^2=(m+1)^2+(n+1)^2$	M1
	$10=(m+1)^2+\dfrac{1}{9}(m+1)^2$	M1
	$9=(m+1)^2$	M1
	$m=-4 \quad n=0$	A1 A1 (6)
ALT (c)	Using Vectors	
	$\overrightarrow{AB}=\begin{pmatrix}7\\21\end{pmatrix}$ so perpendicular to $\overrightarrow{AB}=\begin{pmatrix}21\\-7\end{pmatrix}$	{M1}
	$\left\lvert\overrightarrow{AB}\right\rvert=7\sqrt{10} \Rightarrow, \left\lvert\overrightarrow{AP}\right\rvert=3\sqrt{10}$	{M1,M1}
	$\overrightarrow{PQ}=\dfrac{\sqrt{10}}{7\sqrt{10}}\times\begin{pmatrix}21\\-7\end{pmatrix}=\begin{pmatrix}3\\-1\end{pmatrix}$	{M1}
	So $Q=(-1-3,-1--1)$	{A1}
	$Q=(-4,0)$	{A1}
(d)(i)	$AB=\sqrt{(3+4)^2+(11+10)^2}=7\sqrt{10}$	M1
	$RQ=\sqrt{(-11+4)^2+(-21)^2}=7\sqrt{10}$	A1
(d)(ii)	Gradient of $RQ=\dfrac{-21-0}{-11+4}=3$	M1
	So Gradient of AB (=3) = Gradient of RQ	A1 (4)

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ALT (d)	Using Vectors $\overrightarrow{RQ} = \begin{pmatrix} -4-(-11) \\ 0-(-21) \end{pmatrix} = \begin{pmatrix} 7 \\ 21 \end{pmatrix}$	{M1} {A1}
	$\overrightarrow{AB} = \begin{pmatrix} 7 \\ 21 \end{pmatrix} = \overrightarrow{RQ}$	{M1}
	Because the vectors are the same they must be parallel and the same length	{A1}
(e)	Area $= 7\sqrt{10} \times \sqrt{10} = 70$	M1 A1 (2)
ALT (e)	Using Vectors $\frac{1}{2}\begin{vmatrix} 3 & -4 & -11 & -4 & 3 \\ 11 & 10 & -21 & 0 & 11 \end{vmatrix}$	{M1}
	$= 70$	{A1}
		Total is 16 marks

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Part	Mark	Guidance
(a)	M1	For a fully correct method of finding an equation of a straight line. $\frac{y-y_1}{y_2-y_1}=\frac{x-x_1}{x_2-x_1}\Rightarrow\frac{y-(-10)}{11-(-10)}=\frac{x-(-4)}{3-(-4)}$ Or finds gradient $\frac{11+10}{3+4}=3$ and uses $y+10=3(x+4)$ or $y-11=3(x-3)$ If $y=mx+c$ is used, they must find a complete equation for this mark. Allow **one** error only for the award of this mark.
	A1	For a correct line in any form. $y+10=3(x+4)$ or $y-11=3(x-3)$ or $y=3x+2$ or even $\frac{y+10}{21}=\frac{x+4}{7}$ but do not allow incomplete processing.
(b)	M1	For one correct from $x=-1$ or $y=-1$
	A1	For the correct coordinates of point P $(-1,-1)$ Accept $x=-1$ $y=-1$
(c)	M1	Uses the perpendicular gradient to set up an equation in m and n. $-\frac{1}{'3'}=\frac{n-'(-1)'}{m-'(-1)'}\Rightarrow-\frac{1}{'3'}(m+1)=n+1$ or $n=-\frac{1}{3}m-\frac{4}{3}$ Ft their gradient in part (a) and their P from part (b) for this mark.
	M1	Uses Pythagoras theorem to set up an equation in m and n. $\left(\sqrt{10}\right)^2=(m-'(-1)')^2+(n-'(-1)')^2$ Ft their coordinates of point P form part (b) for this mark.
	M1	Attempts to solve their two equations in n and m simultaneously and forms a quadratic equation in one variable only. $10=(m+1)^2+\frac{1}{9}(m+1)^2\Rightarrow 9=(m+1)^2$ or $0=m^2+2m-8$ or $10=9(n+1)^2+(n+1)^2\Rightarrow 0=10n^2+20n$
	M1	For solving their either: $9=(m+1)^2\Rightarrow m=...$ or $0=10n^2+20n\Rightarrow n=...$ which must be a quadratic equation.
	A1	For finding **either** $m=-4$ **or** $n=0$ Condone the sight of $m=2$ for this mark.
	A1	For finding **both** $m=-4$ **and** $n=0$ $\Rightarrow(-4,\ 0)$ The final answer must be given as coordinates.
	ALT – using vectors – see main scheme.	
(d)(i)	M1	For finding **either** the length $AB=\sqrt{(3+4)^2+(11+10)^2}=7\sqrt{10}$ **Or** $RQ=\sqrt{(-11+4)^2+(-21)^2}=7\sqrt{10}$
	A1	For finding **both** the length $AB=\sqrt{(3+4)^2+(11+10)^2}=7\sqrt{10}$ **And** $RQ=\sqrt{(-11+4)^2+(-21)^2}=7\sqrt{10}$ **and** states they are equal
(d)(ii)	M1	The gradient of $RQ=\frac{-21-'0'}{-11-'(-4)'}='3'$ Ft their coordinates from part (c)

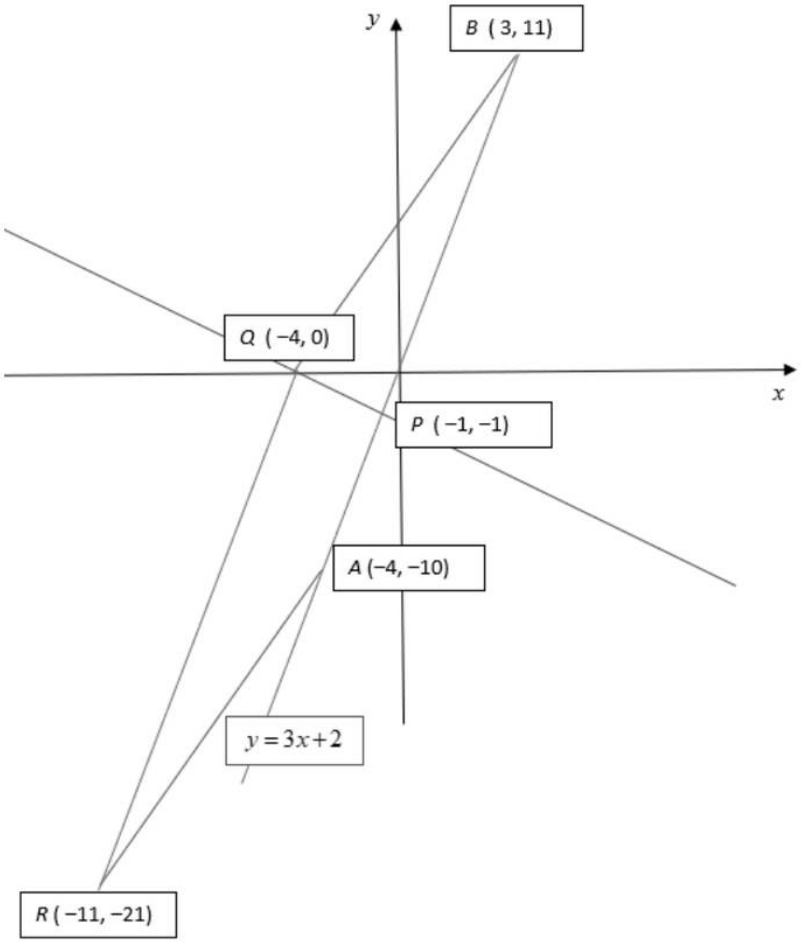
Type MS · Writing the equation of a line through two points

Answer 6 of 12 · Question 4 part a · 2 marks

2021 JUN · Paper 02 · extract 4 of 5 · find part a in this mark scheme extract

	A1	States that the gradient of RQ = gradient of AB [from (a)]
	ALT – Uses vectors, see main scheme. Ft their coordinates of Q – (m, n)	
(e)	**M1**	For a correct expression for the area using their length of AB and the given length of PQ $\left(\sqrt{10}\right)$ Area = $'7\sqrt{10}'\times\sqrt{10}=...$
	A1	For the area = 70 [square units]
	ALT – Uses the discriminant	
	M1	For a correct expression of the area in sequential order using their coordinates for Q Area = $\frac{1}{2}\begin{vmatrix} 3 & -4 & -11 & '-4' & 3 \\ 11 & 10 & -21 & '0' & 11 \end{vmatrix}$
	A1	Area = 70 [square units]

Useful sketch



Type MS · Writing the equation of a line through two points

Answer 6 of 12 · Question 4 part a · 2 marks

2021 JUN · Paper 02 · extract 5 of 5 · find part a in this mark scheme extract

Question number	Scheme	Marks

Type MS · Writing the equation of a line through two points

Answer 7 of 12 · Question 2 part a.i · 0 marks

2020 JAN · Paper 02 (regional R) · find part a.i in this mark scheme extract

The mark scheme extract for this part is not available yet. Use "Check my answer" on the website.

Type MS · Writing the equation of a line through two points

Answer 8 of 12 · Question 2 part a.ii · 3 marks

2020 JAN · Paper 02 (regional R) · find part a.ii in this mark scheme extract

The mark scheme extract for this part is not available yet. Use "Check my answer" on the website.

Type MS · Writing the equation of a line through two points

Answer 9 of 12 · Question 8 part a · 3 marks

2023 JUN · Paper 02 (regional R) · extract 1 of 5 · find part a in this mark scheme extract

8(a)	$\frac{y-5}{5-9}=\frac{x-1}{1-9}$ $\Rightarrow x-2y+9=0$	M1A1 A1 [3]
(b)	Coordinates of point X $\left(\frac{3\times9+1\times1}{3+1},\frac{3\times9+1\times5}{3+1}\right)=(7,8)$ The perpendicular gradient $=-\frac{1}{\frac{1}{2}}=-2$ Equation of l $y-8=-2(x-7)\Rightarrow y=-2x+22$ *	B1B1 B1ft M1A1 cso [5]
(c)	$y=-2(6)+22\Rightarrow p=10$	B1 [1]
(d)	$\overrightarrow{BA}=\begin{pmatrix}1-9\\5-9\end{pmatrix}=\begin{pmatrix}-8\\-4\end{pmatrix}\Rightarrow \overrightarrow{CD}=\begin{pmatrix}6-8\\10-4\end{pmatrix}=\begin{pmatrix}-2\\6\end{pmatrix}$ and coordinates of C are (6, 10) so coordinates of D are $(-2, 6)$	M1 A1A1 [3]
(e)	Length of AB (or CD) $=\sqrt{(9-5)^2+(9-1)^2}=\sqrt{80}$ Length of $CX=\sqrt{(10-8)^2+(7-6)^2}=\sqrt{5}$ Area of parallelogram $ABCD=\sqrt{5}\times\sqrt{80}=\sqrt{400}=20$ (units^2)	B1 B1 M1A1 [4]
		Total 16 marks

Part	Mark	Notes
(a)	M1	For using a correct method and the given coordinates of A and B to form the equation of AB. Do not score this mark until they find the gradient and form the equation of the line using a correct formula. If they use $y=mx+c$ do not allow this mark until they find c and form a complete equation. $\left[m=\frac{1}{2},\quad c=\frac{9}{2}\quad y=\frac{x}{2}+\frac{9}{2}\right]$
	A1	For the correct equation of AB in any form.
	A1	For the correct equation of AB in the required form.

Type MS · Writing the equation of a line through two points

Answer 9 of 12 · Question 8 part a · 3 marks

2023 JUN · Paper 02 (regional R) · extract 2 of 5 · find part a in this mark scheme extract

(b)	B1	For either x or y correct coordinates of point X **NB This is an M mark in Epen**
	B1	For both x and y correct coordinates of point X **NB This is an A mark in Epen.**
	B1ft	For writing down the inverse reciprocal of their gradient for AB Ft their gradient form part (a)
	M1	For forming an equation for l using their coordinates of X and their negative reciprocal gradient of AB $y-8=-2(x-7)\Rightarrow y=-2x+22$
	A1	For the correct equation of l as shown.
(c)	B1	For $y = 10$
(d)	M1	For a suitable method. **Method 1 - Uses vectors:** $\overrightarrow{BA}=\begin{pmatrix}1-9\\5-9\end{pmatrix}=\begin{pmatrix}-8\\-4\end{pmatrix}\Rightarrow\overrightarrow{CD}=\begin{pmatrix}6+(-8)\\10+(-4)\end{pmatrix}\Rightarrow\left[\begin{pmatrix}-2\\6\end{pmatrix}\right]$ OR: $\overrightarrow{BC}=\begin{pmatrix}6-9\\9-10\end{pmatrix}=\begin{pmatrix}-3\\-1\end{pmatrix}\Rightarrow\overrightarrow{AD}=\begin{pmatrix}1-3\\5-(-1)\end{pmatrix}\Rightarrow\left[\begin{pmatrix}-2\\6\end{pmatrix}\right]$ **Method 2 - Uses simultaneous equations**: The equation of AD is $y=-\frac{x}{3}+\frac{16}{3}$ and of CD is $y=\frac{x}{2}+7$ $-\frac{x}{3}+\frac{16}{3}=\frac{x}{2}+7\Rightarrow x=-2$ and $y=6$ **Method 3 - Uses the gradient and length of *AB* or *CD*** $AB=CD=4\sqrt{5}=\sqrt{(6-x)^2+(y-10)^2}$ Gradient: $\frac{1}{2}=\frac{10-y}{6-x}\Rightarrow x=2y-14$ $\left(4\sqrt{5}\right)^2=(6-(2y-14))x^2+(y-10)^2\Rightarrow 5y^2-100y+420=0$ $\Rightarrow y=6,14$ and $x=-2,14$ Allow no more than one error in either method.
	A1	For either correct x or y coordinate of D [(– 2, 6)] **NB** this is an A mark in Epen
	A1	For both correct coordinates of D (– 2, 6)

Type MS · Writing the equation of a line through two points

Answer 9 of 12 · Question 8 part a · 3 marks

2023 JUN · Paper 02 (regional R) · extract 3 of 5 · find part a in this mark scheme extract

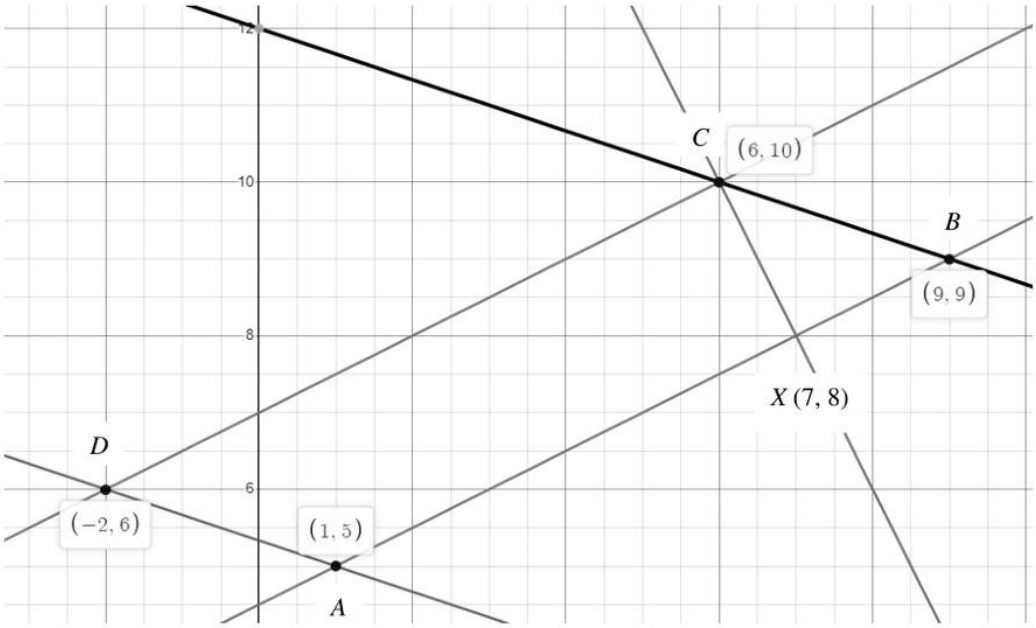
(e)	B1	For the correct length of either ***AB*** **[*CD*] or *CX*** These are given coordinates there is no ft
	B1	For the correct lengths of both ***AB*** **[*CD*] and *CX*** Ft their C and their X
	M1	For using a correct method to calculate the area of a parallelogram. $\text{b} \times \text{h} = CX \times AB = \sqrt{5} \times \sqrt{80} = \sqrt{400} = 20$ If their lengths are incorrect allow this mark provided their lengths are identified as base and perpendicular height. If they use AD as the base $\left[\sqrt{10}\right]$, the perpendicular height required is $2\sqrt{10}$ so base × height = $\sqrt{10} \times 2\sqrt{10} = ...$
	A1	For the correct area of 20 [units^2]
	ALT – Using determinants	
	B1B1	For sight of the correct array which must use the coordinates of A, B, C and D **ONLY.** The coordinates (7, 8) seen in the array is B0B0 $\begin{pmatrix} 1 & 9 & '6' & '-2' & 1, \\ 5 & 9 & '10' & '6' & 5 \end{pmatrix}$ Award both marks for fully correct. Award B1 if there are no more than 2 errors but with none missing. There must be 5 sets of coordinates in the array with first and last the same. **The coordinates must go in order around the parallelogram clockwise or anticlockwise.**
	M1	For the correct evaluation of their 2×5 array. Allow this even if they have (7, 8) included instead of (9, 9) [which is a common error] $\frac{1}{2}\begin{pmatrix} 1 & 9 & 6 & -2 & 1 \\ 5 & 9 & 10 & 6 & 5 \end{pmatrix} = \frac{1}{2}\left\lvert\left[(9+90+36-10)-(45+54-20+6)\right]\right\rvert = ...$
	A1	For the correct area of 20 [units^2]

Type MS · Writing the equation of a line through two points

Answer 9 of 12 · Question 8 part a · 3 marks

2023 JUN · Paper 02 (regional R) · extract 4 of 5 · find part a in this mark scheme extract

Useful sketch



Type MS · Writing the equation of a line through two points

Answer 9 of 12 · Question 8 part a · 3 marks

2023 JUN · Paper 02 (regional R) · extract 5 of 5 · find part a in this mark scheme extract

Question	Scheme	Marks

Type MS · Writing the equation of a line through two points

Answer 10 of 12 · Question 11 part a · 2 marks

2021 NOV · Paper 02 · extract 1 of 3 · find part a in this mark scheme extract

11 (a)	$\frac{y-5}{5--5}=\frac{x-7}{7--3}$ $[\Rightarrow y=x-2]$ oe	M1A1 [2]
(b)	$m_{BC}=\frac{5-1}{7-p}=\frac{4}{7-p}$, $m_{AC}=\frac{1-(-5)}{p-(-3)}=\frac{6}{p+3}$ $m_{BC}\times m_{AC}=-1\Rightarrow\frac{4}{7-p}\times\frac{6}{p+3}=-1\Rightarrow\frac{4}{7-p}=-\left(\frac{p+3}{6}\right)$ $\Rightarrow -24=-p^2+4p+21\Rightarrow p^2-4p-45=0$ $\Rightarrow (p-9)(p+5)=0\Rightarrow p=-5,\ (9)$	M1A1A1 M1 dM1 M1A1* [7]
(c)	Using formula; $\left(\frac{(7\times-5)-(3\times7)}{7-3},\frac{(7\times1)-(3\times5)}{7-3}\right)=(-14,-2)$	B1B1 [2]
(d) (i)	$AC=\sqrt{(1-[-5])^2+(-5-[-3])^2}=\sqrt{40}=2\sqrt{10}$	B1
(ii)	$BD=\sqrt{(7-[-14])^2+(5-[-2])^2}=\sqrt{490}=7\sqrt{10}$ Area $=\frac{1}{2}\times2\sqrt{10}\times7\sqrt{10}=70\ (\text{units}^2)$	M1 M1A1 [4]
		Total 15 marks

Type MS · Writing the equation of a line through two points

Answer 10 of 12 · Question 11 part a · 2 marks

2021 NOV · Paper 02 · extract 2 of 3 · find part a in this mark scheme extract

Part	Mark	Notes
(a)	M1	For forming an equation of a straight line
	A1	Accept a correct line in any form following simplification of denominators; for example, accept $\frac{y-5}{10}=\frac{x-7}{10}$ or $y-5=x-7$ or $y+5=x+3$
(b)	M1	For attempting to find an expression for the gradient of BC **or** the gradient of AC $m_{BC}=\frac{5-1}{7-p}$ or $m_{AC}=\frac{1-(-5)}{p-(-3)}$ Accept maximum one slip in either expression for this mark.
	A1	For an expression for the gradient of BC **or** AC fully correct
	A1	For an expression for the gradient of BC **and** AC fully correct Need not be simplified in either case
	M1	For setting their gradient of BC = negative reciprocal of their gradient of AC
	dM1	For attempting to form a 3TQ in p: $\left[p^2-4p-45=0\right]$ Dependent on the previous M mark.
	M1	For solving their 3TQ by any method. See General Guidance.
	A1*	For $p=-5$
(c)	B1	For either $x=-14$ or $y=-2$
	B1	Both coordinates correct $(-14,-2)$ given as coordinates only.
(d)(i)	B1	For the correct length of AC $\left[=2\sqrt{10}\right]$
(ii)	M1	For the correct method to find length BD $\left[=7\sqrt{10}\right]$ Ft their $x=-14$ or $y=-2$
	M1	For a correct method using $A=\frac{1}{2}\times\text{base}\times\text{height}$ using their AC and BD
	A1	For the correct area of $70\ \left(\text{units}^2\right)$
	ALT – Uses determinants	
	M1	For stating $A=\frac{1}{2}\begin{pmatrix}-3 & 7 & -14 & -3\\ -5 & 5 & -2 & -5\end{pmatrix}$ oe
	M1	For evaluating their determinant $A=\frac{1}{2}\left(\left[-3\times5+7\times-2+-14\times-5\right]-\left[7\times-5+-14\times5+-3\times-2\right]\right)$ oe
	A1	Area = $70\ \left(\text{units}^2\right)$ Which must be a positive value

Type MS · Writing the equation of a line through two points

Answer 10 of 12 · Question 11 part a · 2 marks

2021 NOV · Paper 02 · extract 3 of 3 · find part a in this mark scheme extract

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Type MS · Writing the equation of a line through two points

Answer 11 of 12 · Question 2 part b · 3 marks

2025 JUN · Paper 01 · extract 1 of 5 · find part b in this mark scheme extract

2(a)	$\sqrt{(8-3)^2+(3-2)^2}$ or $\sqrt{(4-3)^2+(7-2)^2}$ or $\sqrt{(8-4)^2+(3-7)^2}$ $AB=AC=\sqrt{26}$ hence isosceles	M1 A1*cso [2]
(b)	Midpoint $M=(6,5)$ or gradient of $AM=1$ e.g. $y-2=\left[\frac{5-2}{6-3}\right](x-3)$ or $y-2=\text{"1"}\times(x-3)\Rightarrow y=x-1$	B1 M1A1 [3]
(c)	Equation of AC $y-2=\left(\frac{7-2}{4-3}\right)(x-3)\Rightarrow[y=5x-13]$ --- Gradient of AB $m=\frac{3-2}{8-3}\left[=\frac{1}{5}\right]$ Gradient of DB $m_p=-\frac{1}{\text{'1/5'}}[=-5]$ Equation of BD $y-3=-5(x-8)\Rightarrow[y=-5x+43]$ $5x-13=\text{'}-5x+43\text{'}\Rightarrow x=\ldots.$ or $y=\ldots$ $\left(\frac{56}{10},15\right)$ oe	M1A1 M1 M1 M1A1 dM1 A1 [8]
ALT I (c) Last 6 marks	$D(x,\text{'}5x-13\text{'})$ $BD^2=(x-8)^2+(5x-13-3)^2$ or $AD^2=(x-3)^2+(5x-13-2)^2$ $26+(x-8)^2+(5x-13-3)^2=(x-3)^2+(5x-13-2)^2$ $26-16x+64-160x+256=-6x+9-150x+225\Rightarrow x=\ldots$ $\left(\frac{56}{10},15\right)$ oe	M1 M1 M1A1 dM1 A1

Type MS · Writing the equation of a line through two points

Answer 11 of 12 · Question 2 part b · 3 marks

2025 JUN · Paper 01 · extract 2 of 5 · find part b in this mark scheme extract

ALT II (c)	$y-2=\left(\frac{7-2}{4-3}\right)(x-3)\Rightarrow[y=5x-13]$	
	or $\overrightarrow{AC}=\begin{pmatrix}1\\5\end{pmatrix}$ oe	M1A1
	$\left('4\sqrt{2}'\right)^2='\sqrt{26}'^2+'\sqrt{26}'^2-2\times'\sqrt{26}'\times'\sqrt{26}'\cos\angle BAC\Rightarrow\cos\angle BAC=...$	M1
	or $\sin\frac{\angle BAC}{2}=\frac{'2\sqrt{2}'}{'\sqrt{26}'}$	
	$\cos\angle BAC=\frac{5}{13}$ or $\angle BAC=67.4$(awrt)	M1
	$\frac{\sqrt{26}}{k\sqrt{26}}=\frac{5}{13}$ or $\cos 67.38...=\frac{\sqrt{26}}{k\sqrt{26}}$	M1
	$k=2.6$ exact	A1
	$3+1\times'2.6'=...$ and $2+5\times'2.6'=....$ or $\sqrt{(x-3)^2+('5x-13'-2)^2}='2.6\sqrt{26}'\Rightarrow x=...,y=...$	dM1
	$(5.6,15)$	A1
		Total 13 marks

Type MS · Writing the equation of a line through two points

Answer 11 of 12 · Question 2 part b · 3 marks

2025 JUN · Paper 01 · extract 3 of 5 · find part b in this mark scheme extract

Part	Mark	Additional Guidance
(a)	M1	For correct use of Pythagoras' theorem to find the length / length^2 of one of the sides. Must have a subtraction of x coordinates and a subtraction of y coordinates with appropriate substitution into Pythagoras' theorem. Or at least one correct vector for AC, AB, BC , or the other way around
	A1	For showing **both** AB and AC have length $=\sqrt{26}$ or length$^2=26$ with conclusion that the triangle is isosceles. e.g. stating AC=AB Correct vectors for AB and AC, followed by $\lvert AC\rvert=\lvert AB\rvert$ oe Allow statements such as 'hence shown' or symbols to indicate conclusion of the demonstration such as QED, // or □.
(b)	B1	For finding the midpoint M of line BC Or finds the correct gradient of $AM=1$
	M1	For finding the equation of the line AM in any form using any correct valid method: e.g. $\begin{cases} 3m+c=2 \\ '6'm+c='5' \end{cases} \Rightarrow m=...,c=...\Rightarrow y=...$
	A1	For $y=x-1$
(c)	M1	For correct method of finding the equation of line AC
	A1	For the correct equation of AC.
	M1	For attempting to find the gradient of AB
	M1	For negative reciprocal of **their** gradient AB
	M1	For attempting to find the equation of the line perpendicular to AB and passing through B, their gradient of BD might not be $-\frac{1}{m_{AB}}$, but must be a changed gradient
	A1	For the correct equation of BD.
	dM1	For attempting to solve their line BD and AC simultaneously. Minimum required for a correct method is to obtain an equation in either x or y (by elimination method or substitution method) and finds a value for x or y Dependent on the previous M mark only
	A1	For (5.6, 15) oe, accept given as $x=5.6$ $y=15$
ALT I (c) Last 6 marks	M1	Use the fact that D is on the line AC, sets up the coordinate of $D(x,'5x-13')$ or $D(x,y)$ where y is replaced by their $'5x-13'$ later
	M1	For correct attempt to find the length/length^2 expression of BD or AD using $D(x,'5x-13')$ or $D(x,y)$
	M1	For correct use of Pythagoras' theorem to set up equation $\lvert AB\rvert^2+\lvert BD\rvert^2=\lvert AD\rvert^2$ in terms of x only
	A1	Correct equation, simplified/ unsimplified
	dM1	For expanding and solving their equation to find $x=\ldots$ Dependent on the previous M mark only
	A1	For (5.6, 15) oe accept given as $x=5.6$ $y=15$

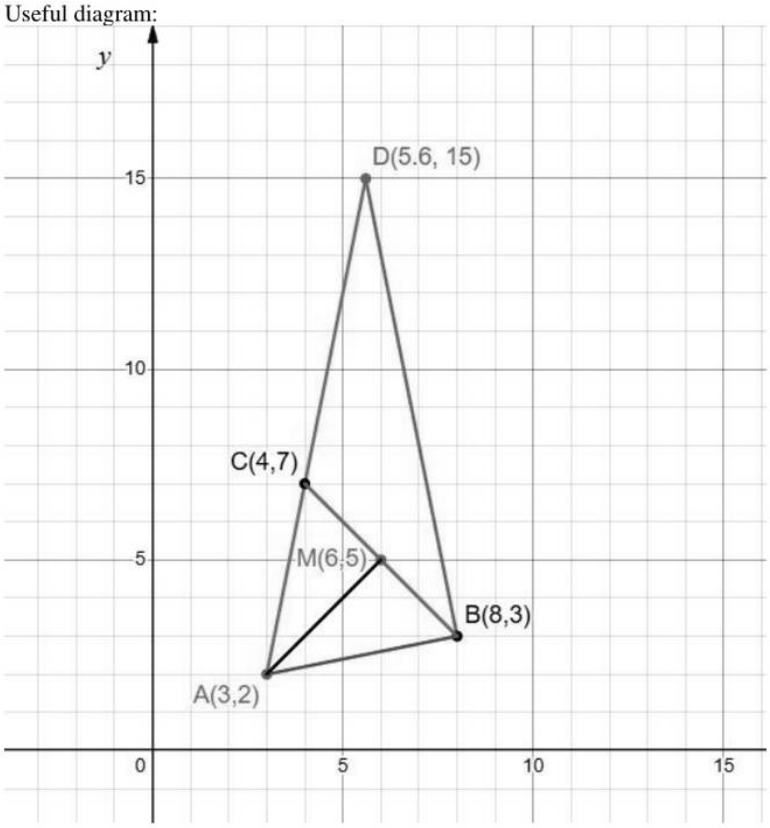
Type MS · Writing the equation of a line through two points

Answer 11 of 12 · Question 2 part b · 3 marks

2025 JUN · Paper 01 · extract 4 of 5 · find part b in this mark scheme extract

ALT II (c)	M1	Correct method of finding the equation of line AC or Vector AC
	A1	Correct equation of AC or correct vector AC
	M1	Correct use of cosine rule or trigonometry to set up an equation involving angle BAC
	M1	Obtains a value of $\cos\angle BAC$ or $\angle BAC$ from a correct cosine rule or trigonometry equation
	M1	Correct equation in k
	A1	Correct k value, must be exact/rounded to 2.6
	dM1	Uses their k value to find $x =\ldots$ and $y =\ldots$ Dependent on the previous M mark only
	A1	Correct coordinates from an exact value of k found, not a rounded value

Useful diagram:



Type MS · Writing the equation of a line through two points

Answer 11 of 12 · Question 2 part b · 3 marks

2025 JUN · Paper 01 · extract 5 of 5 · find part b in this mark scheme extract

Question number	Scheme	Marks

Type MS · Writing the equation of a line through two points

Answer 12 of 12 · Question 10 part c.ii · 2 marks

2025 NOV · Paper 01 · extract 1 of 4 · find part c.ii in this mark scheme extract

Question	Scheme	Marks
10 (a)	$10-3a+a=0 \Rightarrow 10=2a \Rightarrow a=5$ *	M1A1cso [2]
(b)	$(0,-5)$	B1 [1]
(c) (i)	$A \quad (5,10)$ $C \quad (50,-5)$ $y=3x-5 \Rightarrow m_{L1}=3 \qquad m_{L2}=\dfrac{-5-10}{50-5}=\dfrac{-1}{3}$ $m_{L2}\times m_{L1}=\dfrac{-1}{3}\times 3=-1$	B1 M1A1 [3]
(ii)	$y-(-5)=\dfrac{-1}{3}(x-50) \Rightarrow y=\dfrac{-x}{3}+\dfrac{35}{3}$	M1A1 [2]
(d)	$-3=\dfrac{n-("-5")}{m} \Rightarrow n=-5-3m$ $2250=(50-m)^2+(-5-n)^2$ $2250=(50-m)^2+(-5-(-5-3m))^2=(50-m)^2+(3m)^2$ $2250=2500-100m+10m^2 \Rightarrow 0=25-10m+m^2=(m-5)^2$ $\Rightarrow m=5$ $\Rightarrow n=-5-3\times 5=-20$ **ALT** $-3=\dfrac{n-("-5")}{m} \Rightarrow m=\dfrac{-(n+5)}{3}$ $2250=(50-m)^2+(-5-n)^2$ $2250=\left(50+\dfrac{n+5}{3}\right)^2+n^2+10n+25$ $2250=2500+\dfrac{100(n+5)}{3}+\dfrac{n^2+10n+25}{9}+n^2+10n+25$ $\Rightarrow \quad 0=9(2500-2250)+300(n+5)+10n^2+100n+250$ $=2500+300n+1500+10n^2+100n$ $=n^2+40n+400$ $=(n+20)(n+20)$ $\Rightarrow n=-20$ $\Rightarrow m=\dfrac{-(-20+5)}{3}=5$	M1 M1 M1 M1 A1 A1 [6] [M1] [M1] [M1] [M1] [A1] [A1]
(e)	Correctly uses the formula for a kite using their lengths	M1

Type MS · Writing the equation of a line through two points

Answer 12 of 12 · Question 10 part c.ii · 2 marks

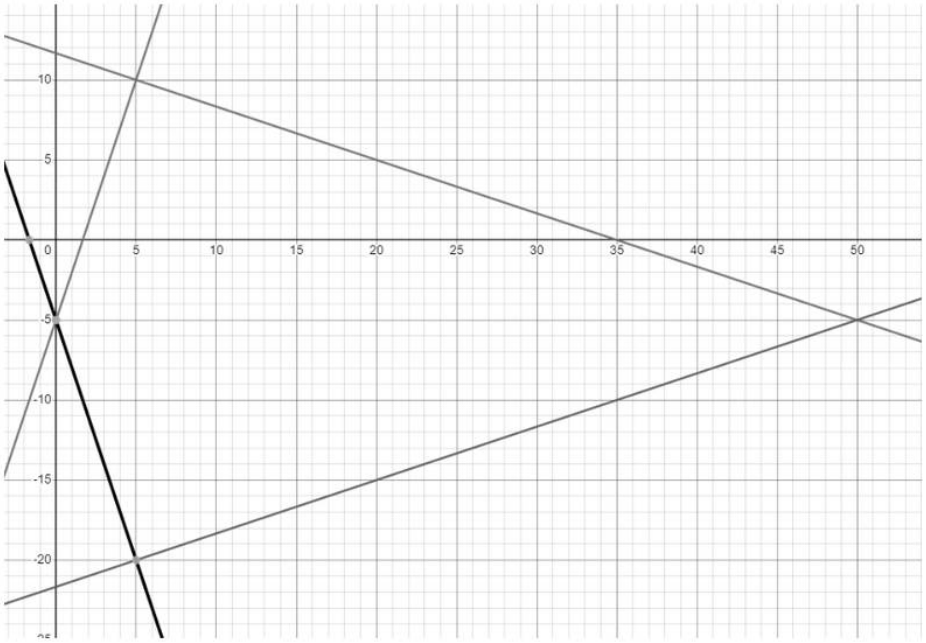
2025 NOV · Paper 01 · extract 2 of 4 · find part c.ii in this mark scheme extract

	Area $= \frac{50\times(10-(-20))}{2} = 750$	dM1A1 [3]
	ALT 1 Correctly uses determinant method with their co-ordinates Area $= \frac{1}{2}\begin{pmatrix} 0 & 5 & 50 & 5 & 0 \\ -5 & 10 & -5 & -20 & -5 \end{pmatrix} = 750$	[M1 dM1A1]
	ALT 2 Correctly uses formulae for two triangles ABD and ACD using their lengths Area $= 2\times\left(\frac{1}{2}\times 15\times 50\right) = 750$ Or Area $= \left(\frac{1}{2}\times 30\times 5\right)+\left(\frac{1}{2}\times 45\times 30\right) = 750$	[M1 dM1A1]
		Total 17 marks

Type MS · Writing the equation of a line through two points

Answer 12 of 12 · Question 10 part c.ii · 2 marks

2025 NOV · Paper 01 · extract 3 of 4 · find part c.ii in this mark scheme extract



Type MS · Writing the equation of a line through two points

Answer 12 of 12 · Question 10 part c.ii · 2 marks

2025 NOV · Paper 01 · extract 4 of 4 · find part c.ii in this mark scheme extract

Question	Scheme	Marks